REV	ECO#	DESCRIPTION OF REVISION
01		INITIAL RELEASE FOR EVT BENJAMIN_MORSE 02/14/25
02	0063087225	UPDATED MM SPEC BENJAMIN_MORSE 04/17/25

NOTES: (UNLESS OTHERWISE SPECIFIED)

1. GEOMETRIC DATA FILES PROVIDED BY APPLE INC. SHALL BE USED FOR MASTER DATA. A DRAWING OF EQUAL FILENAME AND REVISION TO THE GEOMETRIC DATA SHALL BE USED TO DETERMINE GEOMETRIC CHARACTERISTICS AND TOLERANCES. DRAWING DIMENSIONS ARE SUBORDINATE TO GEOMETRIC DATA UNLESS SPECIFIED ON DRAWING TO BE MASTER, AS MAY BE THE CASE OF CABLE LENGTHS OR TABULATED VARIABLE DIMENSIONS. THE SUPPLIER MUST COMPLY WITH THE STANDARDS PERTAINING TO THE TOLERANCE METHOD APPLIED.
2. GD&T - GEOMETRIC TOLERANCE METHODS IN ACCORDANCE TO ASME Y14.5-2009. DATUM REFERENCE FRAMES RANK AS DATUM A PRIMARY, B SECONDARY AND C TERTIARY. FEATURE CONTROL FRAMES MAY RE-ORDER DATUMS PER THE FEATURES DATUM DEPENDENCE. DIMENSIONS THAT LOCATE GEOMETRIC TOLERANCE CHARACTERISTICS ARE BASIC (EXACT). NO TOLERANCES APPLY TO BASIC DIMENSIONS.
3. DIRECT TOLERANCE DIMENSIONING - INTERPRET DIMENSIONS THAT DO NOT LOCATE GD&T CHARACTERISTICS OR GD&T CONTROLLED FEATURES AS THE DIRECT TOLERANCE DIMENSION SYSTEM. THE LOCAL PLUS+/MINUS-, LOCAL LIMIT AND TITLE BLOCK TOLERANCES APPLY.
4. IN THE ABSENCE OF DRAWING DIMENSIONS AND DATUMS, THE GEOMETRIC DATA IS BASIC. THE PLANES OF THE ABSOLUTE COORDINATE SYSTEM DENOTE THE DATUM REFERENCE PLANES AS X-Y = DATUM A, Y-Z = B AND X-Z = C. ALL GEOMETRIC DATA SURFACES ⌓ 0.2 A B C.
5. PERFECT ORIENTATION AND/OR PERFECT LOCATION AT MMC AS REQUIRED FOR THE INTERRELATIONSHIP OF ALL DATUM FEATURES OF SIZE.
6. ON PARTS SUBJECT TO FREE STATE VARIATION, ALL TOLERANCES APPLY WHEN THE PART IS RESTRAINED ACCORDING TO FUNCTIONAL MATING CONDITIONS AS DEFINED BY THE DATUM REFERENCE FEATURE RANK OR FEATURE CONTROL FRAME DATUM RANK PER THE FEATURE INSPECTED. CONSULT PROPER APPLE INC. QUALITY ENGINEER (MQE) FOR APPROVAL/CLARIFICATION ON INDIVIDUAL PARTS.
7. TOLERANCES DESIGNATED WITH THE SYMBOL SPC SHALL BE PRODUCED WITH STATISTICAL TOLERANCE PROCESS CONTROLS. IN ORDER TO USE THE TOLERANCE ON A FEATURE, THE STATISTICAL PROCESS (CP, CPK, ETC) OF THE PART MUST BE VALIDATED AND APPROVED BY AN APPLE INC. MQE.
8. ALL TOOLING, FIXTURING AND OTHER UNIQUE ITEMS THAT ARE USED TO CREATE THIS PART ARE THE PROPERTY OF APPLE INC. AND SHALL BE PERMANENTLY MARKED IN ACCORDANCE WITH DOCUMENT SRAS-2018.
9. THE DESIGN OF ALL TOOLING OR FIXTURING REQUIRED FOR THE MANUFACTURING OR VERIFICATION OF THE PART SHALL BE APPROVED BY THE APPROPRIATE APPLE INC. ENGINEER PRIOR TO TOOL OR FIXTURE FABRICATION.
10. ALL HOMOGENEOUS MATERIALS MUST COMPLY WITH THE FOLLOWING ENVIRONMENTAL SPECIFICATIONS:
    * APPLE INC. REGULATED SUBSTANCES SPECIFICATION, 069-0135
    * ALL ADHESIVES, COATINGS AND PAINTS, PRINTING INKS, AND CLEANING AGENTS USED IN THE MANUFACTURING OF THIS PART MUST COMPLY WITH THE APPLE VOC SPECIFICATION, 099-22549
    * ALL MATERIALS WITH RECYCLED OR RENEWABLE CONTENT MUST COMPLY WITH THE APPLE RECYCLED & RENEWABLE MATERIAL SPECIFICATION, 099-15583
11. MAGNETIC MATERIAL: NdFeB N42SH-100RC (REE) - TO BE APPLE PD/MDE/MQE/GSM APPROVED MATERIAL SUPPLIER MATERIAL PROPERTIES (REPORT PER BLOCK):
    * RESIDUAL MAGNETIC FLUX (Br): 12.8 - 13.2 kGs FAI 21
    * B-COERCIVITY FORCE (Hcb): 12.0 kOe MIN FAI 22
    * J-COERCIVITY FORCE (Hcj): 20.50 kOe MIN FAI 23
    * MAX ENERGY PRODUCT: 40 - 43 MGOe FAI 24
    * BLOCK NdFeB HARDNESS: 600 ± 100 HV FAI 20

[12] MAGNET FINISH: CuNiP PLATING - FOLLOW APPLE PLATING ERS 099-39881
* PHOS CONTENT 9-14% BY WEIGHT
* NO VISIBLE GRAIN OR MACHINING MARKS AFTER PLATING
* GLOSS AT 60°: 177.5 ± 87.5 GU MEASURE AND REPORT 10 PCS PER BATCH FAI 25
* ROUGHNESS: Ra-X / Ra-Y = 0.30 ±0.125 µm Ra FAI 26
* COLOR:
    * L: 60-75
    * a: 1.2-2.7 FAI 27
    * b: 4.5-10

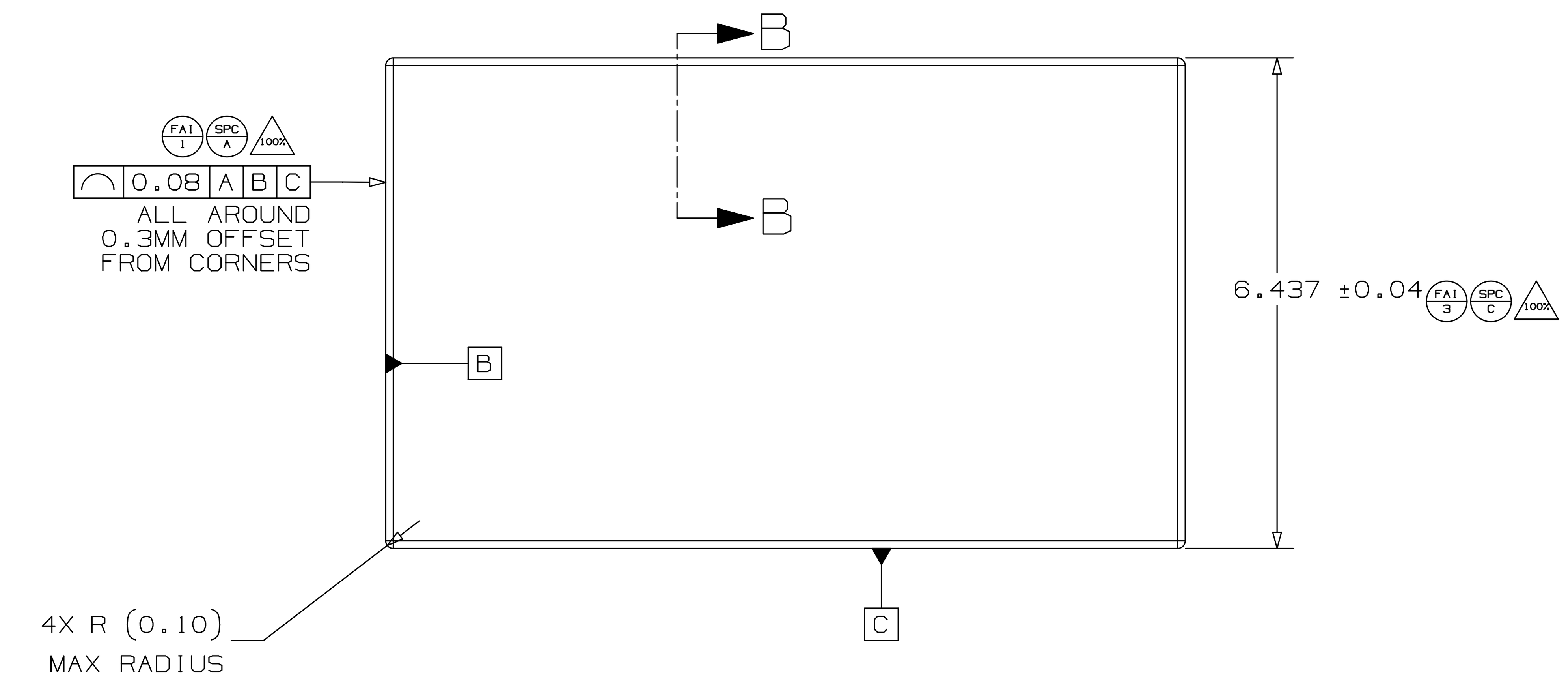
13. NO CRACKS OR CHIPPING AFTER PLATING - CLASS B PER 069-8198 100% APPLE PD/MQE APPROVED COSEMTIC LIMITS
14. PART MUST SURVIVE 24HR SALT SPRAY PER ASTM B117-11. FAI 28 MEASURE AND REPORT 32PCS PER BATCH. NO SIGNS OF CORROSION OR PITTING AFTER PLATING.
15. MAGNETS TO BE SHIPPED NON MAGNETIZED IN APPLE PD/AME/MQE APPROVED PACKAGING
16. DIMENSIONS AFTER PLATING, MAGNET SPECIFIC DIMENSIONS BEFORE PLATING LISTED
17. MAGNETIC MOMENT: 2.910 ± 0.230 uWb-cm MEASURE ON LE FLUXMETER WITH HELMHOLTZ COIL OR OTHER APPLE PD/MDE APPROVED TEST SETUP FAI 29 SPC 0

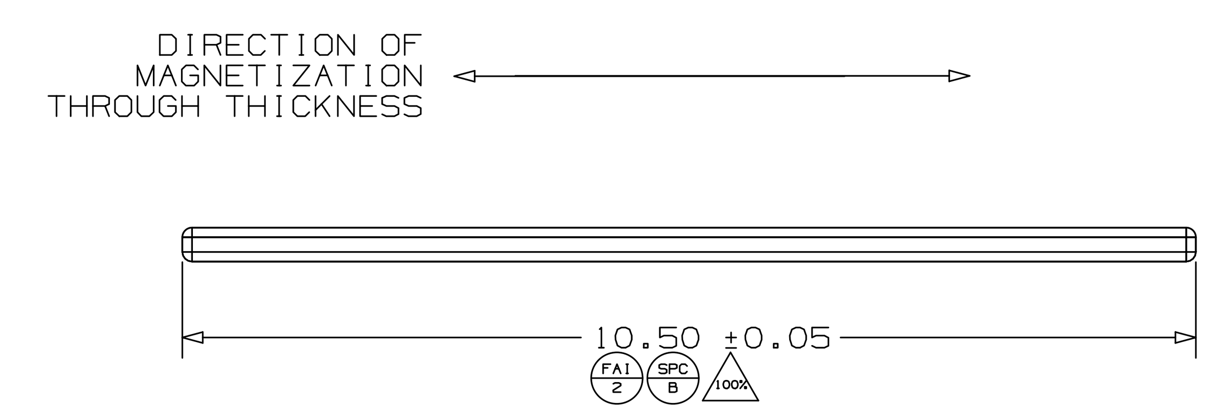
[18] 3PT BENDING STRENGTH:
FORCE TO FAILURE: 30 MIN
DISPLACEMENT TO FAILURE: 0.25 MIN
MEASURE AND REPORT 32 PCS PER BATCH USING APPLE PD/MQE APPROVED TEST SETUP AND PARAMETERS FAI 30

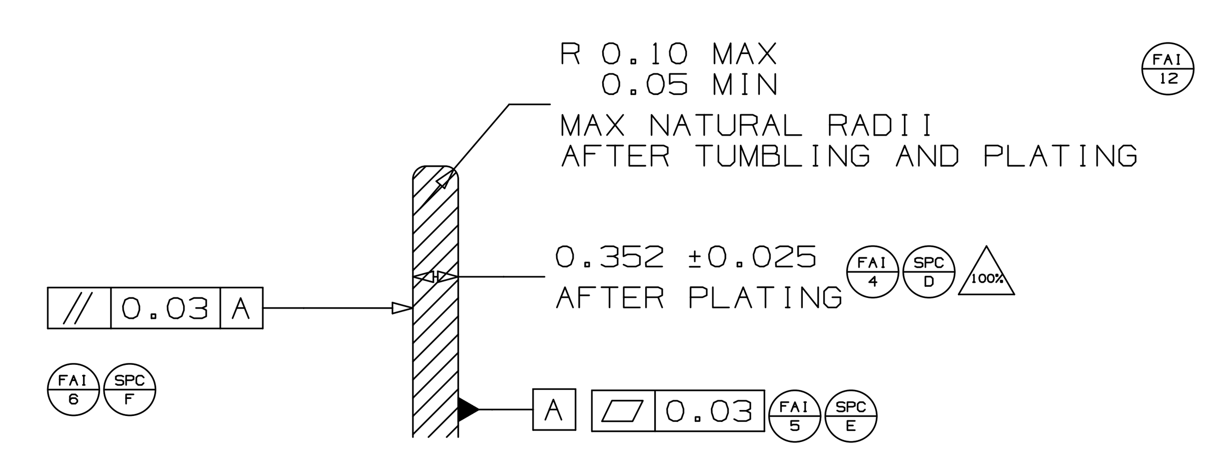
19. PARTS TO SURVIVE 72HR 65° C / 90% RH HEAT SOAK (APPLE 080-1654-K) AND 65° C / 90% RH TO -20° C / UNC TEMPERATURE HUMIDITY CYCLING (APPLE 080-1656-H) FAI 31 WITH LESS THAN 5.0% DECREASE IN MAGNETIC MOMENT. MEASURE AND REPORT 32 PCS PER BATCH USING APPLE PD/MQE TEST SETUP.

[20] MEASURE SURFACE ENERGY USING DYNE PEN ON INDICATED SURFACE.
ACCEPTABLE SURFACE CONDITION AT TO: 32mN/m OR HIGHER USING CONTINUOUS LINE ACROSS PART.
ACCEPTABLE SURFACE CONDITION POST 28 DAY STORAGE DWELL: 30mN/m OR HIGHER USING CONTINUOUS LINE ACROSS PART. FAI 32

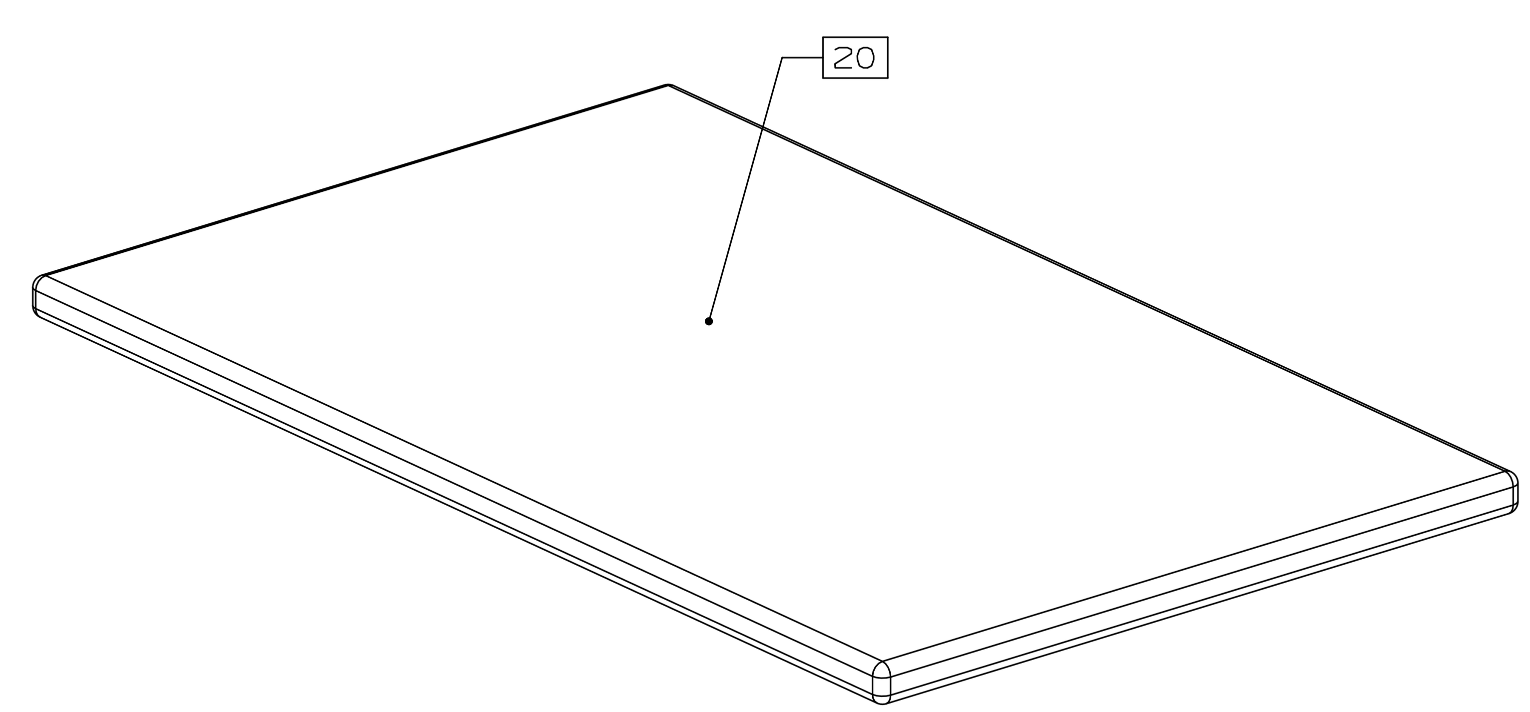
21. 160-05337 TO FOLLOW APPLE MAGNET ERS 080-02981
22. MEASURE WEIGHT (WITHIN 0.01 GRAM RESOLUTION) OF COMPLETED PART ESTIMATED MASS: 0.182 ± 0.014 g FAI 34 DC







SECTION B - B



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DESIGNER: APPLE PD	DATE: 06/06/24	
DIMENSIONS ARE IN MILLIMETERS TOLERANCES		TITLE: MAGNET,OR,QUAD, N42SH,X3132
X.X ±0.2		
X.XX ±0.10		
X.XXX ±0.050		DRAWING NUMBER: 160-05834 REV. 02
ANGLES ±0.5°		
DO NOT SCALE DRAWINGS		
THIRD ANGLE PROJECTION	SIZE D	SCALE NONE SHT 1 OF 2

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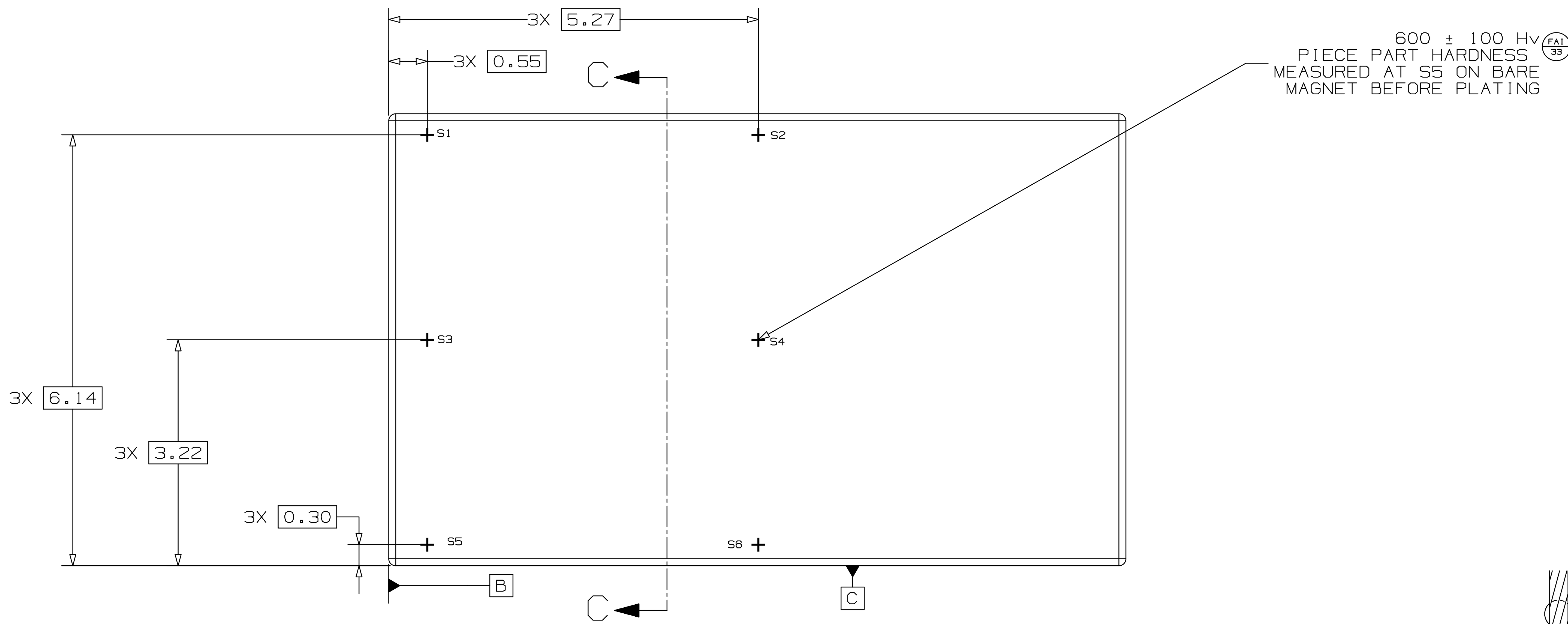
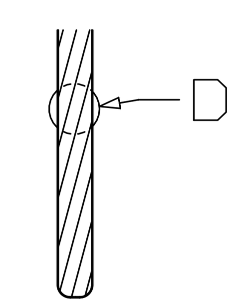
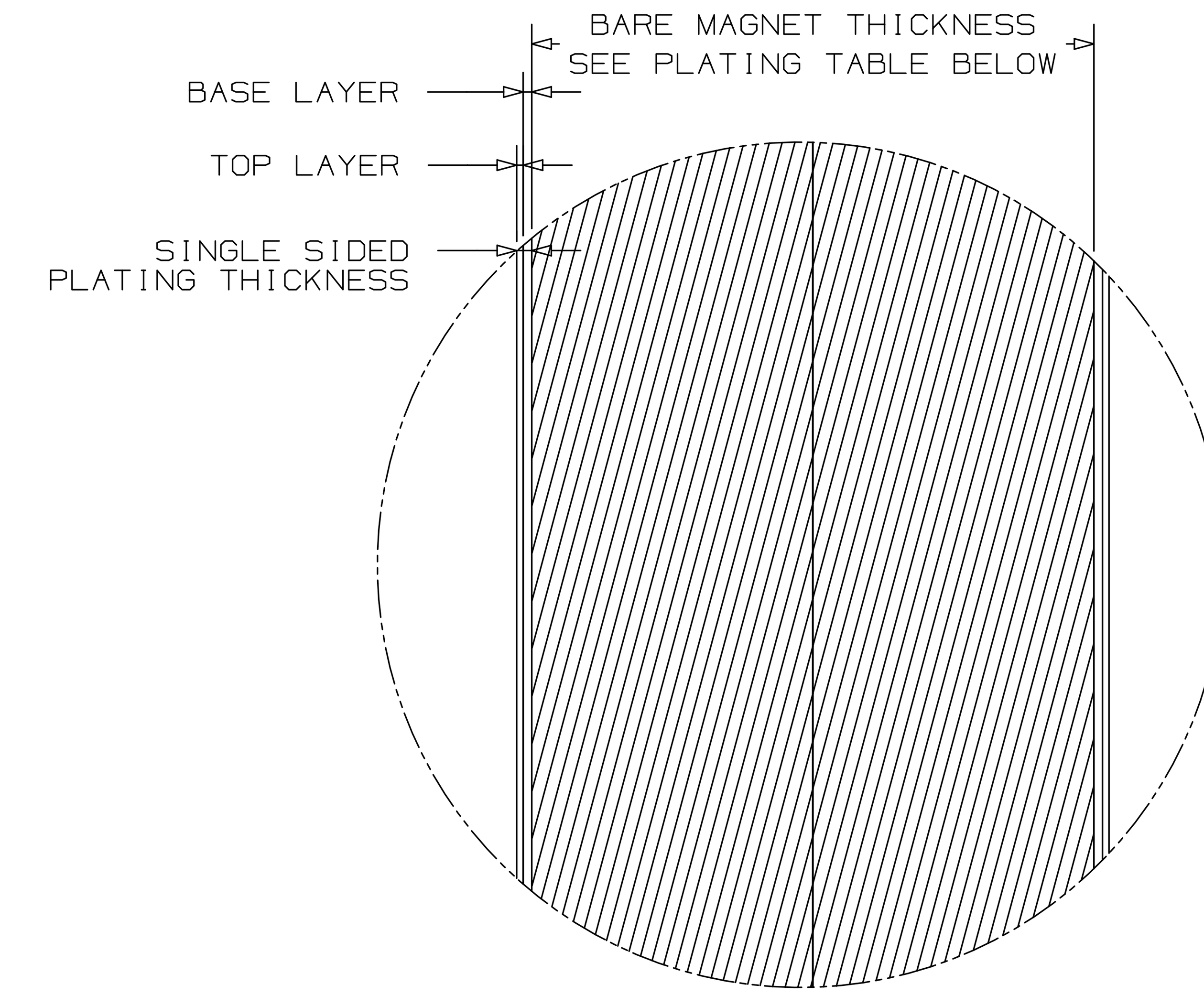


MEASURE PLATING THICKNESS AT 6 LOCATIONS.
MEASURE BARE MAGNET THICKNESS AT S1,S2,AND S4.



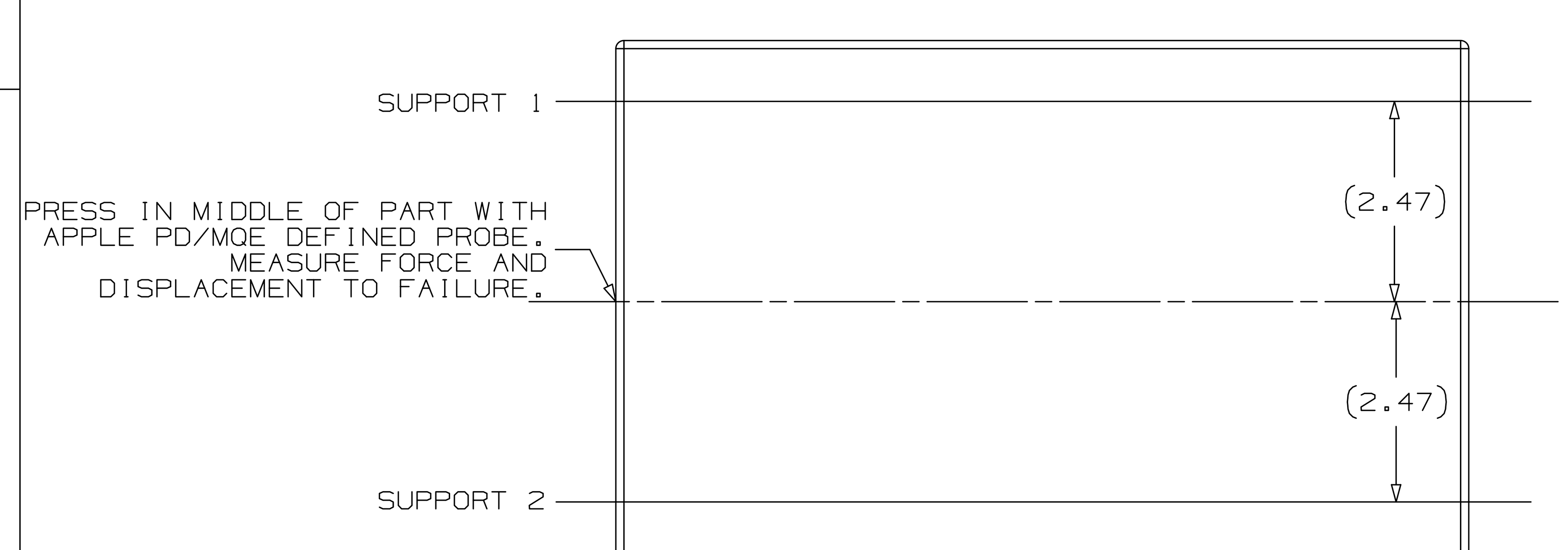
SECTION C - C



DETAIL D

		PLATING SPECIFICATIONS 12	
	FAI 8	TOP LAYER	NiP, 1.00-3.50μm
	FAI 9	BASE LAYER	Cu, 2.00-5.00μm
	FAI 10	SINGLE SIDED PLATING THICKNESS	3.00-8.50μm
11	FAI 7	BARE MAGNET THICKNESS	0.342 ± 0.015

BENDING SUPPORT LOCATIONS 18



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SIZE	DRAWING NUMBER	REV.
D	160-05834	02
SCALE: NONE	SHT 2 OF 2	

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